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Contents

S. No	Title	Page No.
1	Synthesis of silver nanoparticles of some edible basidiomycetes mushroom fungi using response surface methodology and its potential biological application **R Madhanraj, M Eyini and P Balaji**	01
2	Impact of CO_2 on growth, pigments yield and biochemical composition of marine microalga *Dunaliella salina* **A Shenbaga Devi, P Santhanam, S Jeyanthi, B Balaji Prasath and S Dinesh Kumar**	13
3	Fumaronitrile mediated cytochrome P450 (CYP) isoforms biotransformation enzymes responses in *Oreochromis mossambicus* **K Chinnadurai, M Eyini and P Balaji**	23
4	HPLC and biochemical techniques for secondary metabolites in *Garcinia indica* Choisy (Kokum) from transitional zones of Karnataka **Lingappa Sivakumar and Thirugnanasambandam Somasundaram**	35
5	Primary productivity of river chaliyar of Calicut district, Kerala, India **B Dhanalakshmi and P Priyatharsini**	48
6	Anti-bacterial activity, anti-inflammatory and anti- arthritic studies on mangroves by using in vitro model systems **M Babu Selvam and S Abideen**	54
7	Parasitic isopods of the family Cymothoidae from Indian fishes **S Ravichandran and G Ramesh Kumar**	65
8	Isolation and identification of pathogenic bacteria and its antibacterial susceptibility analysis in edible fish *Catla catla* **Mayavan Karthika, Shameem Shabana, Shamoon Muhasin and Venkatachalam Ramasubramanian**	72

9	Biogenic synthesis of silver nanoparticles from *Cardiospermum halicacabum* decorated with Graphene oxide for enhancing antibacterial ability **Gurusamy Sivaprakash, Gujuluva Hari Dinesh, Kulanthaisamy Mohan Rasu, Manoharan Dhivya and Alagarsamy Arun**	80
10	Studies on biosynthesis of xanthan gum using *Xanthomonas* sp., isolated from infected cotton leaves **V Ananthi and A Arun**	88
11	Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample **V Ananthi and A Arun**	95
12	Antibacterial and immunostimulant influence of herbal extracts in grouper *Epinephelus tauvina* experimental culture against *Vibrio harveyi* Infection **T Citarasu, M Michael Babu and SMJ Punitha**	103
13	Assessment of bacteriological quality and presence of antibiotic resistant bacteria in vended sachet-packaged drinking water: potential threat of transmission of enteric pathogens and implications for public health **K Ramamoorthy and Clara G Sargunar**	117
14	Synthesis of chitin form shrimp dispel and its antibacterial activity **P Raja Rajeswari, R Shyamala Gowri, P Meenambigai and K Rajeswari**	132
15	Assessment of antibacterial activity of different solvent extracts of medicinal plant: *Aegle marmelos* **R Shyamala Gowri, R Vijayaraghavan, P Meenambigai and P Raja Rajeswari**	138
16	Effect of aqueous methanolic extract of *Tridax procumbens* on nonspecific immune response of fresh water fish **S Chinniah, T Sangeetha and Subeena Begum**	145
17	A study on biologically synthesize silver nanoparticles using red seaweed *Gracilaria gracilis* **V Veeramanikandan, PT Usha and P Balaji**	154

Characterization and determination of antibacterial activity of bacteriocin producing Lactic acid bacteria isolated from curd sample

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Abstract The purpose of this investigation is to determine the antimicrobial action of *Lactobacillus* delivering bacteriocin isolated from curd sample and to characterize the bacteriocin. Three *Lactobacillus* species were isolated and recognized based on their phenotypic attributes. These strains were able to produce bacteriocin whose antibacterial activity was analyzed by agar well diffusion method against clinical pathogens. The maximum inhibition of growth was observed by the bacteriocin produced by *Lactobacillus acidophilus* against *Staphylococcus aureus* (2.5 mm) followed by *Streptococcus sp*. (2.2 mm) and *Klebsiella pneumonia* (1.4 mm).The growth of *S. aureus* was efficiently inhibited by the other isolates like *Streptococcus lactis* and *Streptococcus cremoris* by forming the zone of inhibition of 2.4 mm and 2.2 mm respectively. The pathogens *Pseudomonas aeruginosa* and *E. coli* were resistant to the isolates producing antibacterial substances. The antibacterial protein bacteriocin was characterized by FT-IR analysis. The isolates demonstrated a productive antibacterial impact against gram positive bacteria. *Lactobacillus acidophilus* strain demonstrated the wide inhibitory range against the gram positive bacteria.

Key Words Bacteriocin, Lactic acid bacteria, FTIR, Inhibitory action.

Introduction

Lactic acid bacteria are a gathering of Gram-positive microbes grouped by morphological, metabolic, physiological attributes (Coeuret *et al*., 2003). They synthesize lactic acid either through homofermentative or heterofermentative pathway and are across the board in nature and furthermore found in human digestive system. *Lactobacilli* are viewed as particularly as gainful microorganisms since they have their capacity to separate proteins, sugars and fats in nourishment and help in assimilation of fundamental components and supplements, for example, minerals, aminoacids and vitamins required for the survival of human beings and other creatures.

The term bacteriocin confers to protein of the colicin sort, portrayed by lethal biosynthesis, intraspecific action and retention to particular receptors and the bacteriocins delivered by *Lactobacillus* fit nearly to the traditional colicin demonstrate (Tagg et al., 1976). A few bacteriocin producing strains have been isolated from crude fermented products (Rodriguez *et al*., 2000).

A few sorts of bacteriocins from food related lactic acid bacteria have been recognized and described, of which the critical ones are acidophilin, nisin, diplococcin, lactacins, bulgarican, diplococcin, helveticins, and plantaricins (Nettles and Barefoot, 1993).

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Among these, bacteriocin and nisin produced by *Lactococcus lactis* sp. *lactis*, has been the most widely described (Buchman *et al*., 1988; Liu and Hansen, 1990).

Lactic acid bacteria apply a huge antagonistic activity against numerous food contaminating microorganisms due to the production of organic acids, hydrogen peroxide, diacetyl, inhibitory chemicals and bacteriocins (Piard and Desmazeaud, 1991). Bacteriocins are ribosomally synthesised antibacterial proteinous substances or peptides, which are delivered by different microscopic organisms for the restriction of development and destruction of closely related bacterial strains.

Bacteriocins are proteinaceous antibacterial compounds and display bactericidal activity against species closely related with the producer strain (De Vugst and Vandamme, 1994) numerous bacteriocins are dynamic against food borne pathogens particularly against *Listeria monocytogenes* (Vignolo *et al*., 1996; De Martins and Franco, 1998; Bredholt *et al*., 1999).

The bacteriocins produced restrained food deterioration and pathogenic bacteria, for example, *Staphylococcus aureus, Eschericia coli, Bacillus cereus, Bacillus subtilis, Listeria monocytogens, Clostridium perfringens* and *Klebsiella*, which are refractory to conventional food conservation method (Bizani and Brandelli., 2002).

Regardless, studies relating to the antibacterial properties of these organisms have been limited and not fully exploited for use (Reddy *et al*., 1984; Abdel-Bar *et al*., 1987). Three of the most essential aspects in the investigation of bacteriocins are their production, characterization and purification. In this way, this paper reports the production and characterization of bacteriocin by lactic acid bacteria isolated from fermented food substances like curd.

Materials and methods

Isolation of Lactic acid bacteria

Packed Curd sample was utilized for the isolation of *Lactobacillus sp*. curd sample was serially diluted and the samples were plated onto De Man Rogosa Sharpe (MRS) medium for *Lactobacillus* isolation. Incubation conditions were 37°C for 48-72 hrs. Three numbers of well isolated colonies with typical characteristics namely pure white, small (2-3mm distance across) with whole edges were picked from each plate and exchanged to MRS broth.

Identification of the Bacterial strains

Advance distinguishing proof of the *Lactobacilli* was performed by their morphological, cultural, physiological and biochemical attributes: (Kandler and Weiss, 1986; Sharpe et al., 1979): Gram reaction, catalase production, carbohydrate fermentation patterns, growth at 15°C and 45°C in the *Lactobacilli* De Mann Rogosa and Sharpe (MRS) broths as depicted by Bergey's Manual of Systematic Bacteriology, resistance of various grouping of Nacl (4 and 6.5%) and Growth on 0.3% Methylene blue. The distinguished variety *Lactobacillus* was further identified to the species level based on their capacity to ferment sugars (Singh and Rakesh., 2009).

Detection of inhibitory action by agar-well diffusion assay

The potential bacteriocin producing strains were identified by agar well diffusion method. The strains were cultured on MRS broth at 37°C for 2 days. Then the cells were pelleted out from the broth by centrifugation at 5000 rpm for 10 min at room temperature.

Around 6mm measurement wells were made on pre inoculated agar media and each well was inoculated with 50 µl of culture supernatant of bacteriocin producing *Lactobacillus* strains after neutralization with NaOH (Toba *et al*., 1991). Inhibitory activity was performed against certain Gram positive and Gram negative organisms: *Streptococcus sp, Staphylococcus aureus, Klebsiella pneumonia, Pseudomonas aeruginosa* and *E.coli*. Hindrance zones around the wells were measured and recorded.

Separation of crude bacteriocin

Lactobacillus species were cultured in 1000 ml MRS broth (pH 7.0, glucose, 0.25% w/v, peptone, 0.5% w/v) for 72 h at 30°C. For extraction of bacteriocin, a cell free supernatant was acquired by centrifuging (10,000 rpm for 20 min. at 4°C) the culture and was acclimated to pH 7.0 by using 1M NaOH to prohibit the antimicrobial impact of organic acids.

Purification of bacteriocin by Ammonium Sulfate Precipitation

The crude bacteriocin samples produced were treated with solid ammonium sulfate up to 60% saturation. The blends were mixed thoroughly for 2 h at 4°C and later centrifuged at 20,000 rpm for 1 h (4°C). The precipitates were re-suspended in 25 ml of 0.05 M potassium phosphate buffer (pH 7.0).

Characterization of bacteriocin by FT-IR analysis

Examination of the bacteriocin produced by all the three lactic acid bacteria was done by FT-IR analysis. The precipitated bacteriocin was blended with KBr and the pelleted out under pressure. The transmittance mode used during analysis was 4000 to 400 cm^{-1}.

Assurance of bacteriocin activity by agar well diffusion assay (Schillinger and Lucke, 1989; Takahiro *et al*., 1991)

Aliquots of 50 µl from each bacteriocin sample were placed in wells in plates seeded with the clinical pathogens (*Streptococcus sp, Staphylococcus aureus, Klebsiella pneumonia, Pseudomonas aeruginosa* and *E.coli)*. The plates were incubated overnight at 30ºC and the zone of inhibition was measured. (Rammelsberg and Radler, 1990).

Results and discussion

Bacteriocins produced by the lactic acid bacterial isolates make them intrigue attributes that legitimize their review. The most striking is that none of these bacteriocins is restricted by the extremely converged antibacterial spectrum reported for some bacteriocins of few lactic acid bacteria, for instance lactococcin A (Holo *et al*., 1991) and lactacin B (Barefoot and Klaenhammer, 1983). Prior reports (Tagg *et al*., 1976; Daeschel *et al*., 1985; Sanni *et al*., 1999) have demonstrated that some bacteriocins produced by gram-positive bacteria have an expanded spectrum of activity. In the current study the *Lactobacillus acidophilis* demonstrated extraordinary range of restraint against the clinical pathogens. The Lactic acid bacteria responsible for bacteriocin production were isolated from the curd sample. About eight isolates were collected based on its distinct colony morphology. Among them only three isolates with bacteriocin producing capability were screened by agar well diffusion method and used for further study (Fig. 1)

The biochemical outcomes demonstrated that every one of the three strains segregated were Gram positive and non-motile. In addition, none of the strain

demonstrated any catalase activity. It was further observed that in 0.3% methylene blue, one of the tests that recognize *Streptococcus lactis* from *Streptococcus cremoris*, while *S. lactis* produce no growth rather than *S. cremoris* shows significant growth on the medium supplemented with 0.3% Methylene blue. Both the strains indicated appropriate growth on 4% NaCl, while no growth was seen in 6.5% NaCl. All the separated strains were further affirmed by sugar tests and the outcomes are exhibited in Table 1 and 2. The outcomes show that *Streptococcus cremoris* indicated positive responses for lactose, sucrose, glucose, maltose, galactose and fructose, with the exception of mannitol. The outcomes for *Streptococcus cremoris* demonstrate that these strains gave positive responses with lactose, glucose, galactose and fructose while gave negative responses with sucrose, maltose, and mannitol. *Lactobacillus acidophilus* gave positive responses with lactose, glucose, maltose, galactose, sucrose and fructose and negative responses with mannitol. These outcomes are similar to the Bergey's manual of determinative bacteriology (Holt et al., 1994). From the biochemical characateristics of the isolates it is clear to affirm them as *Streptococcus lactis*, *Streptococcus cremoris* and *Lactobacillus acidophilus*.

Bacteriocins are cationic hydrophobic anti microbial compounds that are produced by a wide range of species, including LAB (Lopez & Belloso., 2008). Extraction of bacteriocin in this study using Ammonium sulphate indicated that bacteriocin can be precipitated from the cell free supernatant. This suggests that the precipitation happened as a result of increasing hydrophobic interactions of protein molecule, thereby reducing the solubility of the protein.

FTIR investigation of bacteriocin produced from the strains of *Lactobacillus acidophilus*, *Streptococcus lactis* and *Streptococcus cremoris* expressed that it might contain amide I of organic acids and carboxylic groups of bacteriocin. The conspicuous groups of lactic corrosive are the carboxylic groups (OH-C=O) which lies in the range of wave numbers 1760 – 1670 cm^{-1} and at 1460.35 and 1638.90 cm^{-1} uncovered the nearness of amide I . The identification and increment of these peaks in the spectra got by FTIR show the positive production of lactic acid in bacteriocin (Fig. 2-4). An elevation in peak height is straightly corresponding to an increment in amount (Einarsson and Lauzon., 1995).

As indicated by Fricourt *et al.*, (1994), lactic acid bacteria synthesize bactericidal agents that differ in their spectra of activity. A large portion of these specialists are bacteriocins with a proteinaceous dynamic moiety, while others are non-protein operators (Piard and Desmazeaud, 1991, 1992). Analysis of the antimicrobial activity exerted by the bacteriocin was tested against clinical pathogens like *Streptococcus sp*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *E.coli* (Fig. 5). Most extreme zone of inhibition was seen in *Staphylococcus aureus* by bacteriocin produced from *Lactobacillus acidophilus* and followed by *Streptococcus cremoris* and *Streptococcus lactis*.

Since the outer membrane of gram negative microorganisms do not permit particles like bacteriocin to achieve its site of activity, which is the cell membrane (Tukel *et al*., 2007). Additionally in the present study maximum inhibition was seen in the Gram positive organism *Staphylococcus aureus*

instead of the gram negative organism like *Pseudomonas aeruginosa* and *E.coli*. (Fig. 6)

Conclusion

The unique antimicrobial property of *L. acidophilus*, *S. cremoris* and *S. lactis* can have positive impact on the use of respective bacteriocins for traditional fermented foods against food borne pathogens with a view to improving the hygiene and safety of the food products so produced.

Table 1: Biochemical and physiological properties of Lactic acid producing Bacteria

Culture	A	B	C
Gram's staining	+	+	+
Morphology	Rods	Cocci in chains	Cocci in chains
Catalase test	-	-	-
Motility	-	-	-
Growth on 4% NaCl	++	+++	+
Growth on 6.5% NaCl	++	+++	+
Growth at 10°C	-	+	-
Growth at 45°C	+	+	+
Growth on 0.3% Methylene blue	+	-	-
Isolate identified as	***Lactobacillus acidophilus***	***Streptococcus cremoris***	***Streptococcus lactis***

Positive result: (+) Negative result: (-)

Fig. 1: Colonies of Lactic acid bacteria isolated from curd sample

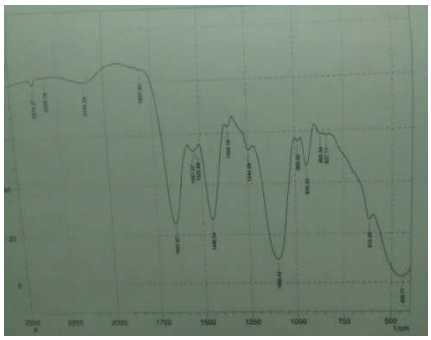

Fig. 2: FTIR spectrum of bacteriocin produced by *Lactobacillus acidophilus*

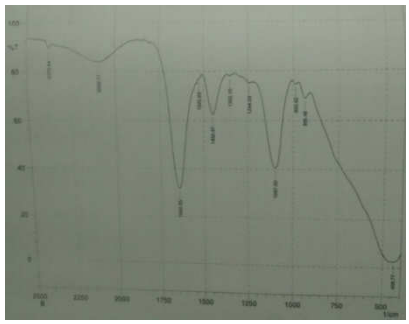

Fig. 3: FTIR spectrum of bacteriocin produced by *Streptococcus cremoris*

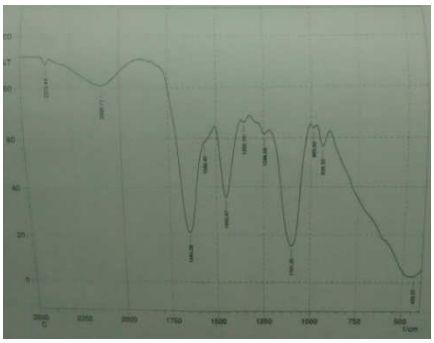

Fig. 4: FTIR spectrum of bacteriocin produced by *Streptococcus lactis*

Fig. 5: Antimicrobial activity of bacteriocin against *Klebsiella pneumonia*

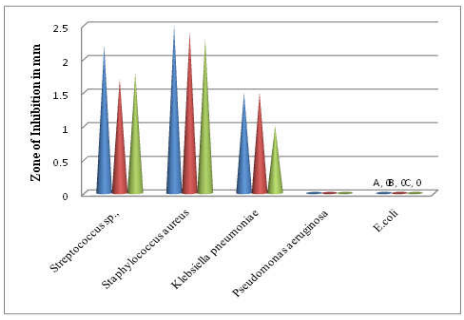

Fig. 6: Antimicrobial activity of bacteriocin against different clinical pathogens

Table 2: Sugar fermentation by Lactic acid bacteria

Culture	Glucose	Sucrose	Lactose	Maltose
A	+	+	+	+
B	+	+	+	-
C	+	+	+	+

Acid production - (+)

No acid production - (-)

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